

THE UNIVERSITY OF AUCKLAND**SEMESTER TWO 2022****MECHENG 726**
Acoustics for Engineers**Time Allowed: 3 hours and 30 additional minutes****NOTE:**

1. This assessment is open book. You are permitted to access your course manuals and other written material including online resources. The assessment must be your own work.
2. The assessment carries a total of 60 marks and is weighted at 60% of your overall mark for this course.
3. It contains 2 questions. Each question has multiple parts. Question 1 contains material from Dr Yusuke Hioka's sections and is worth 20 marks. Question 2 contains material from Dr Michael Kingan's sections and is worth 40 marks.
4. Answer ALL questions including ALL parts of each question.

Submission instructions:

1. Question 2 requires you to submit a single pdf file for each part (i.e. you should submit four separate pdf files in total - one each for questions 2a, 2b, 2c, and 2d). On the first page of your solutions, write the course code (MECHENG 726) and your student ID number.
2. Scan your solutions / working (or export to pdf, if using a digital device) to create a SINGLE pdf file for EACH PART QUESTION (2a, 2b, 2c, or 2d).
3. Check your pdf to make sure all pages are up the right way and in the correct order.
4. Upload a SINGLE pdf of your solutions to EACH PART QUESTION (2a, 2b, 2c, or 2d).

i By submitting this assessment, I agree to the following declaration:

As a member of the University's student body, I will complete this assessment with academic integrity and in a fair, honest, responsible, and trustworthy manner. This means that:

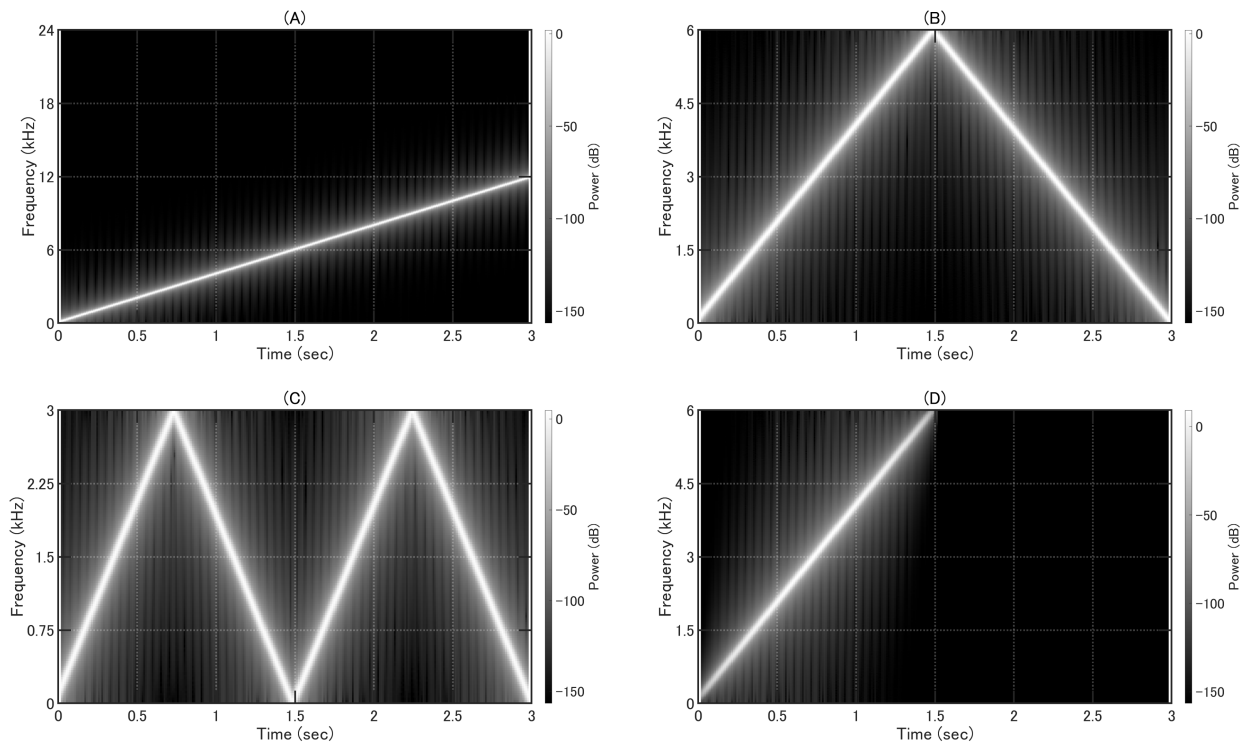
- I will not seek out any unauthorised help in completing this assessment. Unauthorised help includes, but is not limited to, asking another person, friend, family member, third party, tutorial, search function or answer service, whether in person or online.
- I will not discuss or share the content of the assessment with anyone else in any form during the assessment period, including but not limited to, using a messaging service, communication channel or discussion forum, Canvas, Piazza, Chegg, third party website, Facebook, Twitter, Discord, social media or any other channel within the assessment period.
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- I am aware the University of Auckland may use Turnitin or any other plagiarism detecting methods to check my content.
- I declare that this assessment is my own work, except where acknowledged appropriately (e.g., use of referencing).
- I declare that this work has not been submitted for academic credit in this or another University of Auckland course, or elsewhere.

I understand the University expects all students to complete coursework with integrity and honesty. I promise to complete all online assessment with the same academic integrity standards and values.

Any identified form of poor academic practice or academic misconduct will be followed up and may result in disciplinary action.

I confirm that by completing this exam I agree to the above statements in full.

- 1 An acoustical engineer records the sound of the linear swept sine signal (100 Hz to 12 kHz; 3 second length) played from a loudspeaker using a digital sound recorder equipped with a single microphone. Figures (A) to (D) below show the spectrograms of the recorded audio files. Answer the following questions.



- (i) Choose the correct spectrogram of the recorded sound when the sampling frequency of the recorder is set at 48 kHz.

(1 mark)

- ☐ (A)
- ☐ (B)
- ☐ (C)
- ☐ (D)

- (ii) Choose the sampling frequency of the recorder the engineer sets to acquire the signal shown in Figure (B).

(1 mark)

- ☐ 3 kHz
- ☐ 6 kHz
- ☐ 12 kHz
- ☐ 24 kHz
- ☐ 44.1 kHz
- ☐ 48 kHz

(iii) When the signal shown in Figure (D) is recorded by setting the sampling frequency at 12 kHz, choose the function in the recorder the engineer uses.

(1 mark)

- ☐ Enabling a digital low-pass filter (implemented by a digital processor)
- ☐ Enabling Hann window
- ☐ Enabling 24 bit/sample recording
- ☐ Enabling an analogue low-pass filter (implemented by analogue circuit)

(iv) Assume that the linear swept sine signal (100 Hz to 12 kHz; 3 second length) is recorded by pulse code modulation (PCM) with 16 bits/sample and the sampling frequency of the recorder is set at twice the Nyquist rate of the sound. Determine the size of the file saved on the recorder in bytes.

(1 mark)

bytes

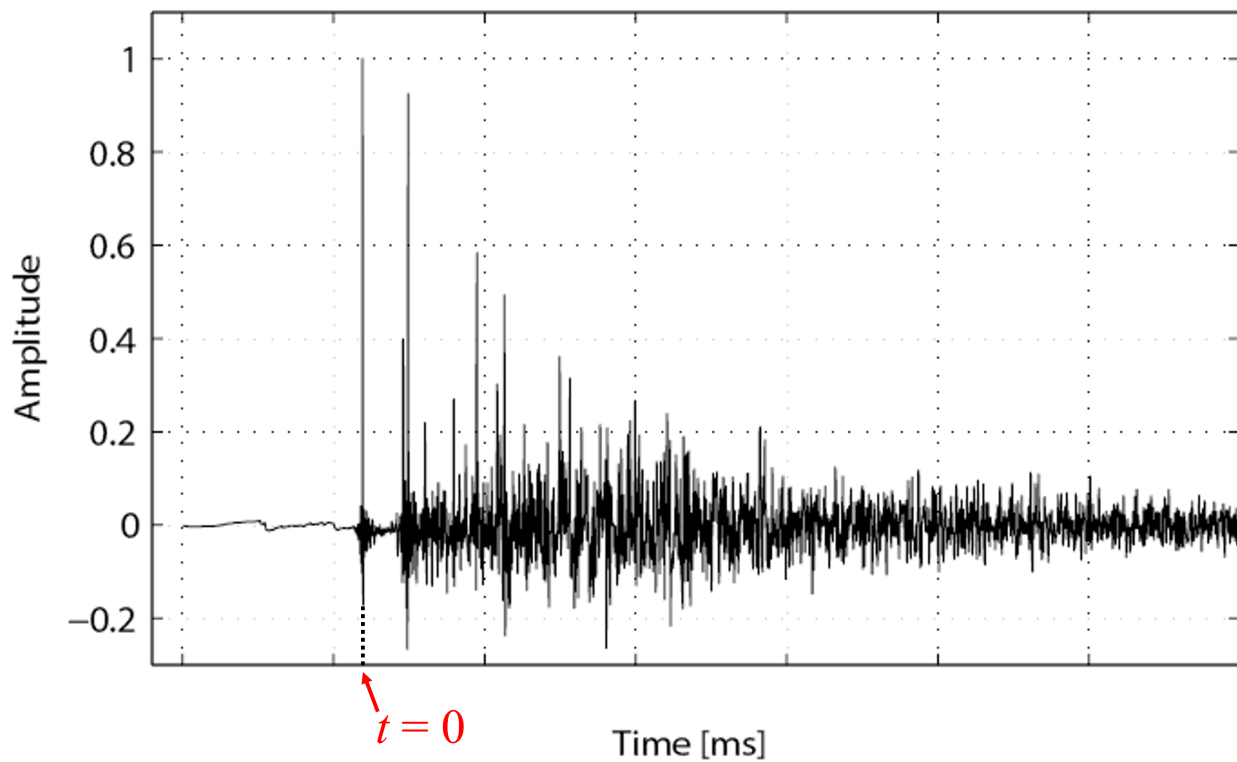
Maximum marks: 4

- 2 An acoustical engineer measured various acoustical properties of a lecture theatre. The following text describes the procedure used for taking the measurements. Fill in all blanks with appropriate words or values.

(7 marks)

"First of all, the room impulse response was measured in the room using a loudspeaker and a calibrated instrumentation microphone, which were located at the centre of the room 2 metres apart from each other. There were no obstacles between the loudspeaker and the microphone (i.e. the loudspeaker was located in the direct line of sight of the microphone). The measured room impulse response was processed to derive the following metrics/properties.

- : The measured room impulse response was split into two parts, and the ratio of their energy was calculated by $10 \log_{10} \frac{\int_{t=0}^{\tau} h^2(t) dt}{\int_{t=\tau}^{\infty} h^2(t) dt}$ to derive this property, where $h(t)$ denotes the measured room impulse response and t is the time. Here the limit of the integrals τ was set as $\tau = 50 \text{ ms}$ provided that $t=0$ was defined at the time when the arrived at the microphone (which is the time of the highest peak in the measured room impulse response shown in the figure below).
- **Reverberation time:** The early decay curve (EDC) was calculated from the room impulse response, and its slope was estimated using linear regression. Because the early decay time (EDT) was used, a linear regression was applied to the time period between two points dB (higher point) and dB (lower point) on the EDC below the initial level of the EDC. The formula given by the linear regression was $y = At + B$, where $A = -120 \text{ dB/s}$, $B = 20 \text{ dB}$ and y is the linear fit to the EDC in dB. Hence, the reverberation time of the room was determined as seconds.
- **Frequency response:** was applied to the measured room impulse response in order to derive the frequency response of the lecture theatre. Because the measured room impulse response also includes the property of the loudspeaker, the frequency response of the loudspeaker separately measured in a/an/the was used to isolate the acoustic properties of the lecture theatre."



Maximum marks: 7

- 3 The following statements are concerned with the physiology of the human auditory perception system. Choose the correct word(s) for each blank to complete the statements.

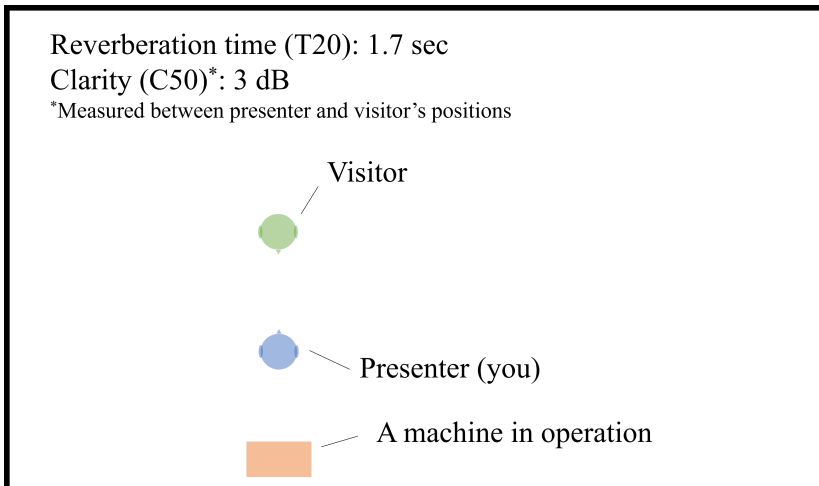
(2 marks)

- i) The (ear canal, cochlea, semicircular canal, pinna, ear drum) is a part of the outer ear which helps us determine which angle/direction a sound source is located at.
- ii) The role of the (outer ear, outer hair cells, pinna, inner hair cells, middle ear, cochlea) is to efficiently transmit the energy of sound from the ear drum to the cochlea.
- iii) The (pinna, outer hair cells, ear drum, ear canal, Inner hair cells) convert the vibration of the (basilar membrane, ear canal, inner hair cells, outer hair cells, ear drum) to neural signals transmitted to the brain.

Maximum marks: 2

- 4 You are giving a presentation in English in a factory to an overseas visitor whose first language is not English, however, you find the visitor struggling with understanding your speech. Discuss ALL factors that are potentially affecting the visitor's understanding and how you could improve the visitor's understanding by referring to the plan and the acoustical characteristics of the space shown in the diagram below.

(4 marks)



Fill in your answer here

Format | **B** | *I* | U | $\times_2$ | $\times^2$ | $\frac{\square}{\square}$ | $\sqrt{\square}$ | $\int \square$ | $\frac{\square}{\square}$ | $\leftarrow$ | $\rightarrow$ | $\curvearrowright$ | $\frac{1}{2}$ | $\frac{3}{4}$ | Ω | $\square$ | $\pencil$ | Σ |

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Words: 0

Maximum marks: 4

5 i) Choose ALL correct statements about loudspeakers.

(1 mark)

- ☐ A power amplifier is always needed to drive the driver of a loudspeaker.
- ☐ Drivers of loudspeakers work based on the same principle as condenser microphones.
- ☐ Drivers of loudspeakers are attached to an enclosure to produce louder sound.
- ☐ 2-way or 3-way loudspeakers have more than one driver to play sound louder.

ii) Choose ALL potential applications of microphone arrays.

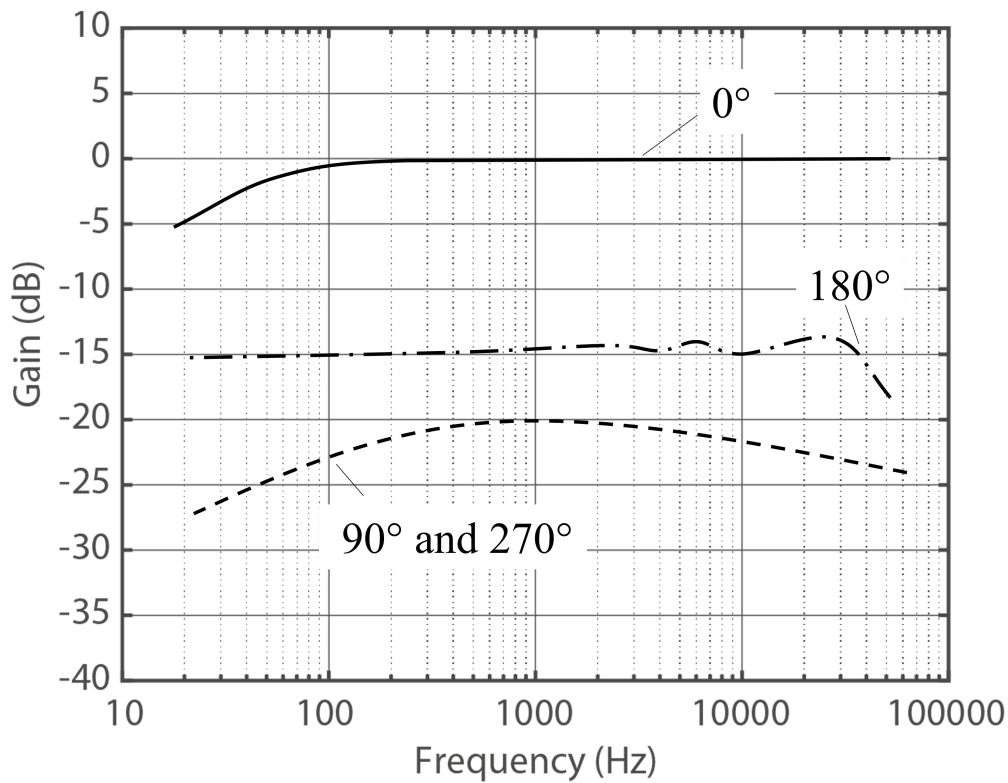
(1 mark)

- ☐ Estimating the location of a sound source.
- ☐ Measuring the acoustical characteristics of target environments that are virtually reproduced by a sound reproduction system.
- ☐ Emphasising sound arriving from a particular direction.

Maximum marks: 2

- 6 The following diagram shows the frequency response of a microphone measured in an anechoic chamber. The annotations indicate the angle of incidence of the sound relative to the microphone axis. The directivity of this microphone is (uni-directional (cardioid), bi-directional (figure of eight), omni-directional, shotgun).

(1 mark)



Maximum marks: 1

7 2a. Workers in a factory are exposed to the following sound levels during their working day:

Time	Level ($L_{A,eq}$)
08:00 – 10:00	90 dB
10:00 – 10:30	60 dB
10:30 – 12:00	92 dB
12:00 – 12:30	62 dB
12:30 – 16:00	88 dB

(i) Calculate a worker's noise dose for this work day. [5 marks]

(ii) A hunter is employed by the Department of Conservation to cull rabbits in the Southern Alps. During the course of his work, the hunter fires his gun which produces a pressure impulse. This pressure impulse is modelled as

$$p = P_0 \exp\left\{-\left(\frac{t}{\tau}\right)^2\right\}$$

where $P_0 = 100$ Pa, and $\tau = 10^{-3}$ s.

Estimate L_{peak} for this impulse. Also comment on whether the hunter should wear hearing protection in order to comply with New Zealand workplace law.

[3 marks]



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Maximum marks: 8

- 8 (i). A simple (monopole) source produces a tone with a frequency of 1000 Hz. The sound power radiating from the source is 1 Watt. The source is located 1 m above an infinitely large rigid plane. Calculate the sound pressure level measured by a microphone lying on the rigid plane at a location 10 m horizontally from the source. Assume that the density of air $\rho_0 = 1.2 \text{ kg/m}^3$, and the speed of sound in air $c = 340 \text{ m/s}$.

[3

marks]

- (ii) For the case considered in part (i), calculate the sound pressure level measured by a microphone located 10 m horizontally from the source and 1 m above the rigid plane.

[7

marks]



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Maximum marks: 10

- 9 Two classrooms are separated by a common wall. The classrooms are identical in shape being rectangular prisms with internal dimensions of 10 m wide $\times$ 20 m long $\times$ 4 m high. The common wall measures 10 m wide $\times$ 4 m high. The reverberation time in both rooms is 1 s.

(i) Calculate the total sound absorption in one of the classrooms.

[2

marks]

(ii) If the maximum expected reverberant sound pressure level in either of the classrooms is 85 dB, calculate the sound reduction index of the common wall required to ensure that the reverberant sound pressure level in the other classroom due to sound transmission is less than 35 dB.

[3

marks]

(iii) If a child can be modelled as a sound source with a sound power level of 70 dB, calculate the reverberant sound pressure level in a classroom containing 20 children.

[5

marks]



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Maximum marks: 10

- 10** Consider an impedance tube with a circular cross section. The tube has a loudspeaker at one end (with the diaphragm located at $x = L$) and a foam sample at the other (with face closest to the speaker located at $x = 0$). The speaker produces sound at a single frequency, i.e. is time-harmonic so that the complex pressure within the tube has the form $P(x, t) = P(x) \exp\{i\omega t\}$.

The acoustic field within the tube can be assumed to be one-dimensional. This field consists of two plane waves – an incident wave with complex amplitude, P_i , travelling towards the sample (in the negative x direction) and a reflected wave with complex amplitude, P_r , travelling away from the sample (in the positive x direction). The complex sound pressure within the tube is given by the following expression.

$$p(x, t) = P_i \exp\{i\omega(t + \frac{x}{c})\} + P_r \exp\{i\omega(t - \frac{x}{c})\}$$

- (i) Show that the pressure amplitude within the tube is given by

$$|p| = |P_i| \sqrt{1 + |R|^2 + 2\Re\{R \exp\{-2i\omega(x/c)\}\}}$$

where $R = P_r/P_i$ is the complex pressure reflection coefficient and $\Re\{z\}$ denotes the real part of a complex number z .

[4

marks]

- (ii) If the speaker produces an incident wave within the tube with a sound pressure level of 40 dB at 1000 Hz, calculate the maximum sound pressure level within the tube if the foam sample has a reflection coefficient $R = 0.3$.

[8

marks]



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Maximum marks: 12